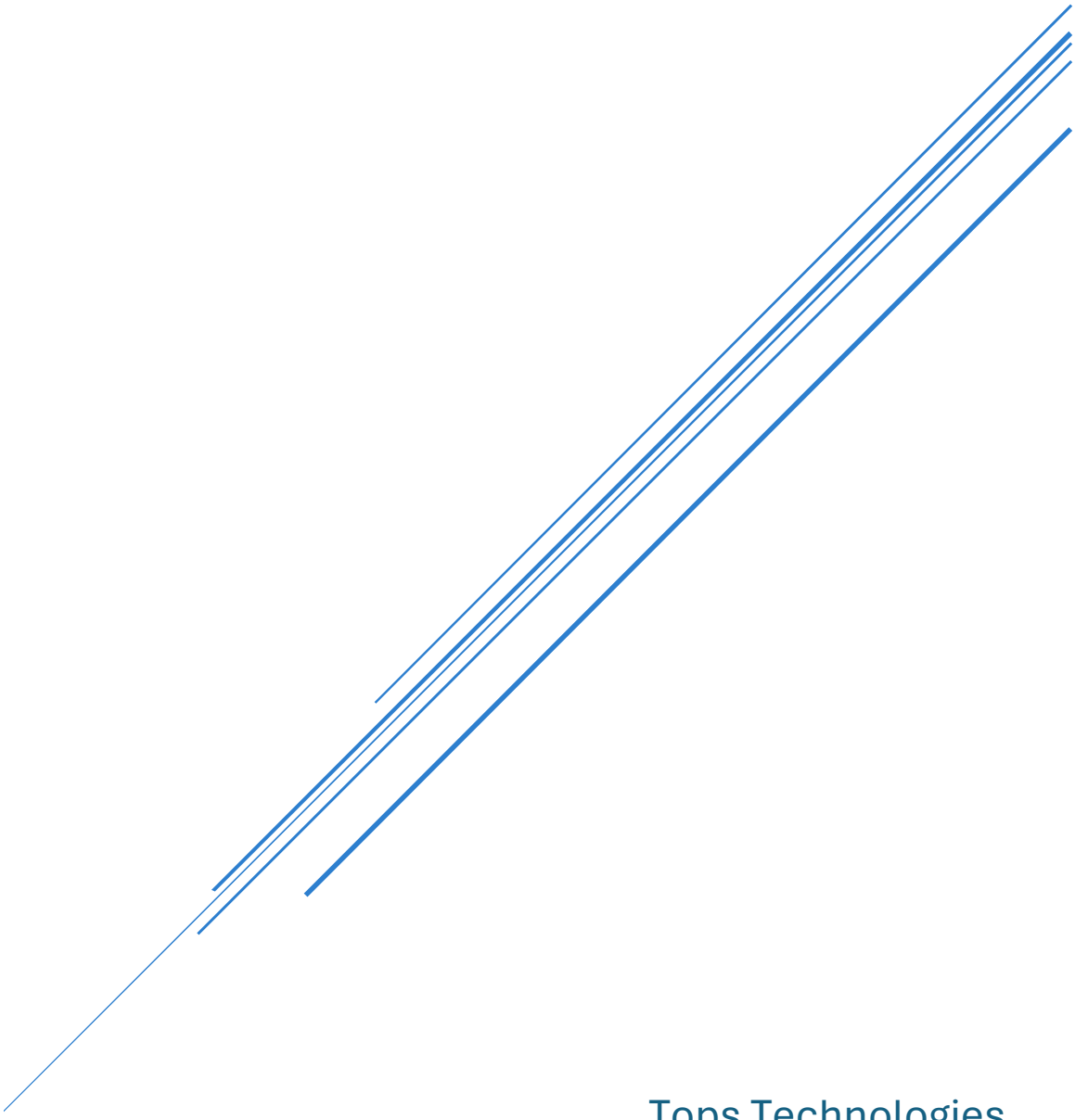
DAY – 6 INTRODUCTION TO NETWORK DEVICES

Assignment - 17



Tops Technologies
Cyber Security

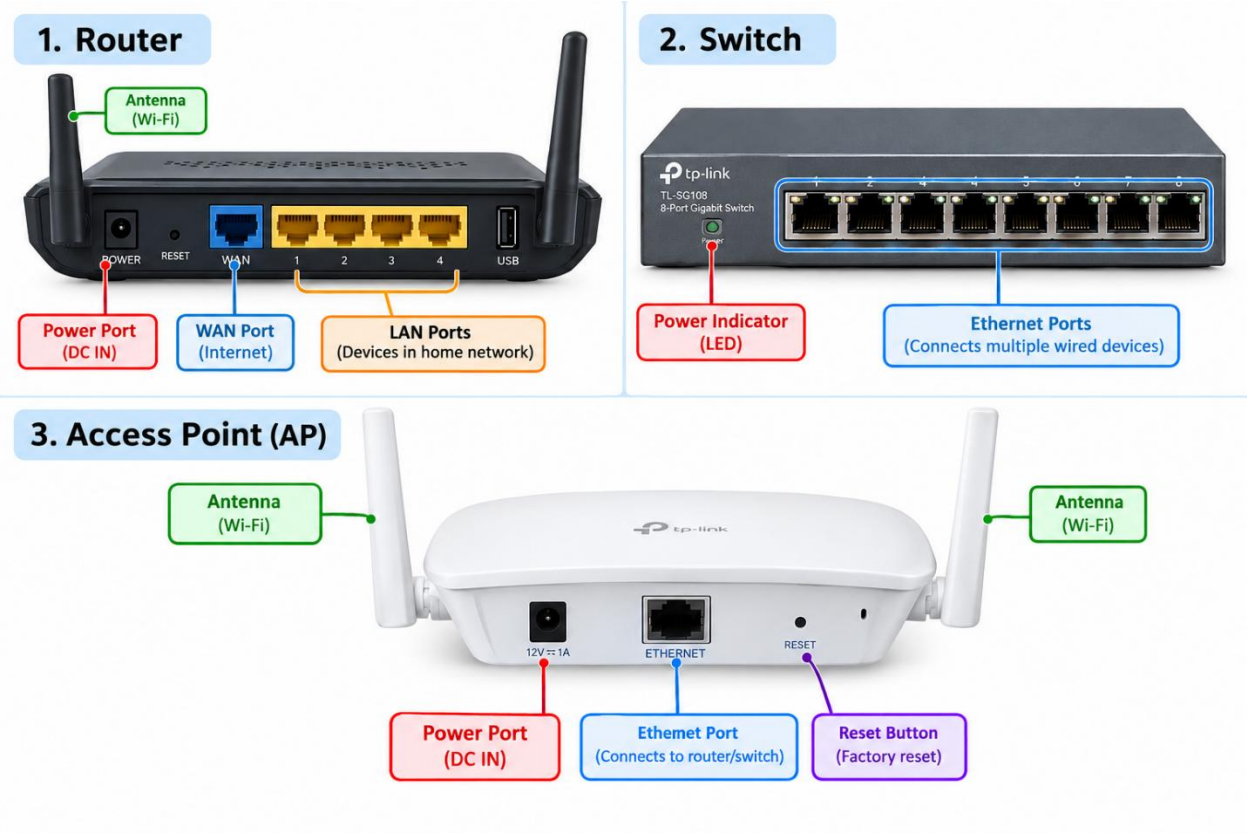
1. **Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.**

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Network Device	Main Function	Typical Ports
Switch	Connects multiple wired devices, such as PCs and printers, within a local network (LAN).	5, 8, 16, or 24 Ethernet ports (RJ-45)
Router	Connects different networks and forwards data between the home network and the internet.	1 WAN port and 2–4 LAN Ethernet ports
Hub	Connects multiple devices and sends incoming data to all connected ports.	4, 8, or 16 Ethernet ports (RJ-45)
Firewall	Protects the network by monitoring and filtering incoming and outgoing traffic according to security rules.	Physical Ethernet ports (WAN/LAN); logical ports such as 80 (HTTP) and 443 (HTTPS)
Access Point (AP)	Provides Wi-Fi connectivity so wireless devices can connect to a wired network.	1 or more Ethernet ports (RJ-45), often with PoE support

2. **Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.**

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1. **Router: Connects the home network to the internet. Its ports include WAN, LAN, and power.**
2. **Switch: Connects multiple wired devices in a LAN using Ethernet ports.**
3. **Access Point: Provides wireless connectivity. It has Wi-Fi antennas, an Ethernet port, and a power port.**

3. **Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.**

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I use Paytm to make digital payments and check my transaction history.

1. **Access Point: It connects my mobile phone to the internet through Wi-Fi.**
2. **Router: It forwards data between my home network and the internet when I use Paytm.**
3. **Firewall: It helps filter unauthorized or suspicious network traffic to improve security.**

4. **Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.**
   Hint: Think about device capacity, coverage, and traffic handling.

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A possible reason for slow college Wi-Fi and frequent disconnections is that the access points are overloaded because many students and staff are connected to the network at the same time. When an access point has limited capacity, it may struggle to handle the traffic from all connected devices. Poor Wi-Fi coverage, outdated routers, or overloaded switches can also affect network performance.

Solution: The college can upgrade its existing access points to modern, high-capacity devices that support more simultaneous connections and provide better wireless coverage. It can also upgrade the router to handle network traffic more efficiently and use high-speed switches to improve communication between wired devices. These upgrades can reduce network congestion, improve internet speed, minimize disconnections, and provide a more reliable Wi-Fi experience for students and staff.